Proxy Methods 1 - Olley-Pakes (1996)

Olley and Pakes (1996) consider the effects of deregulation on productivity, using data on plants in the US telecommunications sector

They consider estimation of a Cobb Douglas production function for (the log of observed) value added

$$\widetilde{v}_{it} = \beta_K k_{it} + \beta_L l_{it} + a_{it} + \varepsilon_{it}$$

where a_{it} is (the log of unobserved) TFP and ε_{it} is an additive measurement error component of $\widetilde{v}_{it}$, which is assumed to be uncorrelated with (k_{is}, l_{is}) for all s, t

$$\widetilde{v}_{it} = \beta_K k_{it} + \beta_L l_{it} + a_{it} + \varepsilon_{it}$$

OP discuss estimation of (β_K, β_L) for the case in which we have quantity data on value added, capital and labour, which we follow here

[In their empirical application, they use deflated value measures of output and capital, and an hours measure of labour]

[OP also allow output to vary with the observed age of plants, and allow for selection biases due to non-random entry and exit of plants, which we abstract from here]

$$\widetilde{v}_{it} = \beta_K k_{it} + \beta_L l_{it} + a_{it} + \varepsilon_{it}$$

OP treat l_{it} as a perfectly flexible input, but assume that k_{it} is chosen by the firm in period $t-1$, after observing $a_{i,t-1}$ and period $t-1$ prices, but before observing a_{it} and period t prices

- this timing assumption plays a key role in their procedure

Capital may also be subject to (strictly convex) adjustment costs

They assume that a_{it} evolves according to a first-order Markov process

$$a_{it} = g(a_{i,t-1}) + \xi_{it}$$

where $g(a_{i,t-1}) = E(a_{it}|\Omega_{i,t-1}) = E(a_{it}|a_{i,t-1})$ and $E(\xi_{it}|a_{i,t-1}) = 0$

Their key assumptions are that the decision rule for investment in period t, or equivalently for capital in period $t+1$, can be written in the form

$$k_{i,t+1} = i_t(k_{it}, a_{it})$$

as a (possibly time-specific) function of the state variables k_{it} and a_{it} *only*, and that $i_t(k_{it}, a_{it})$ is *strictly monotonic* in a_{it}

The time-dependence of $i_t(k_{it}, a_{it})$ allows for common variation over time in the price of capital goods, or other components of the user cost of capital, but this assumption rules out any heterogeneity across firms in a given time period

This assumption also rules out any *serially correlated* variation across firms in wages, but allows for *serially uncorrelated* variation across firms in wages (which is irrelevant for the optimal level of $k_{i,t+1}$ given the maintained timing assumption)

The *strict* monotonicity of $i_t(k_{it}, a_{it})$ rules out any *non-convex* forms of adjustment costs, which imply a region of inaction, in which given k_{it}, optimal investment remains zero for a range of values of a_{it}

Under these assumptions, OP propose a two-stage procedure in which β_L is estimated in the first stage, and β_K is then estimated in the second stage

OP First Stage

Given strict monotonicity, we can invert $k_{i,t+1} = i_t(k_{it}, a_{it})$ to obtain

$$a_{it} = h_t(k_{i,t+1}, k_{it})$$

This allows us to replace the unobserved a_{it} in the production function by a general (possibly time-specific) function of $k_{i,t+1}$ and k_{it} (or, equivalently, by a general (possibly time-specific) function of k_{it} and investment in period t)

$$\begin{aligned}
\widetilde{v}_{it} &= \beta_K k_{it} + \beta_L l_{it} + a_{it} + \varepsilon_{it} \\
&= \beta_K k_{it} + \beta_L l_{it} + h_t(k_{i,t+1}, k_{it}) + \varepsilon_{it} \\
&= \beta_L l_{it} + \phi_t(k_{i,t+1}, k_{it}) + \varepsilon_{it}
\end{aligned}$$

where $\phi_t(k_{i,t+1}, k_{it}) = \beta_K k_{it} + h_t(k_{i,t+1}, k_{it})$

$$\widetilde{v}_{it} = \beta_L l_{it} + \phi_t(k_{i,t+1}, k_{it}) + \varepsilon_{it}$$

We do not identify β_K from this first-stage specification, but we do identify β_L *provided there is serially uncorrelated variation across firms in the real wage* $(w_{it} - p_{it}^V)$ - Ackerberg, Caves and Frazer (2015)

Since l_{it} is assumed to be a perfectly flexible input, we require variation in the real wage to generate identifying variation in l_{it} given k_{it} and a_{it}

Since this wage variation cannot affect the investment decision, it must be serially uncorrelated

We also require a quantity measure of the labour input

$$\widetilde{v}_{it} = \beta_L l_{it} + \phi_t(k_{i,t+1}, k_{it}) + \varepsilon_{it}$$

Under these assumptions, the 'control function' $h_t(k_{i,t+1}, k_{it})$ acts as a proxy for the unobserved a_{it}, and we can estimate β_L consistently from this first-stage specification

If we are willing to approximate $\phi_t(k_{i,t+1}, k_{it})$ using a polynomial in $(k_{i,t+1}, k_{it})$, possibly with time-varying coefficients, then we can estimate β_L consistently using OLS, and inference is standard

Otherwise we can treat $\phi_t(k_{i,t+1}, k_{it})$ non-parametrically, and estimate β_L consistently using variants of Robinson's (1988) semi-parametric estimator for partially linear regression models

OP Second Stage

Given consistent estimates of β_L and $\phi_{it} = \phi_t(k_{i,t+1}, k_{it})$ from the first-stage specification, the second stage of the OP procedure uses

$$\begin{aligned}
\widetilde{v}_{it} - \beta_L l_{it} &= \beta_K k_{it} + a_{it} + \varepsilon_{it} \\
&= \beta_K k_{it} + g(a_{i,t-1}) + \xi_{it} + \varepsilon_{it} \\
&= \beta_K k_{it} + g(h_{t-1}(k_{it}, k_{i,t-1})) + \xi_{it} + \varepsilon_{it} \\
&= \beta_K k_{it} + g(\phi_{t-1}(k_{it}, k_{i,t-1}) - \beta_K k_{i,t-1}) + \xi_{it} + \varepsilon_{it} \\
&= \beta_K k_{it} + g(\phi_{i,t-1} - \beta_K k_{i,t-1}) + \xi_{it} + \varepsilon_{it}
\end{aligned}$$

For example, if we specify

$$\begin{aligned} a_{it} &= g(a_{i,t-1}) + \xi_{it} \\ &= \rho a_{i,t-1} + \xi_{it} \end{aligned}$$

to be a linear AR(1) process, then we have

$$\begin{aligned} \widetilde{v}_{it} - \beta_L l_{it} &= \beta_K k_{it} + g(\phi_{i,t-1} - \beta_K k_{i,t-1}) + \xi_{it} + \varepsilon_{it} \\ &= \beta_K k_{it} + \rho(\phi_{i,t-1} - \beta_K k_{i,t-1}) + \xi_{it} + \varepsilon_{it} \end{aligned}$$

and β_K (and ρ) can be estimated by non-linear least squares, finding the values of (β_K, ρ) which minimize the variance of $(\xi_{it} + \varepsilon_{it})$, after plugging in consistent estimates of β_L and $\phi_{i,t-1}$ obtained from the first stage

$$\widetilde{v}_{it} - \beta_L l_{it} = \beta_K k_{it} + \rho(\phi_{i,t-1} - \beta_K k_{i,t-1}) + \xi_{it} + \varepsilon_{it}$$

Identification of (β_K, ρ) from this second-stage specification *requires* that the variance of $(\xi_{it} + \varepsilon_{it})$ has a *unique minimum* at the true values of those parameters

For the linear AR(1) process for a_{it}, this condition *fails* if the decision rule for $k_{i,t+1} = i_t(k_{it}, a_{it})$ is *an exact log-linear function* of (k_{it}, a_{it})

- Bond and Söderbom (in preparation)

If we have

$$k_{i,t+1} = \theta_K k_{it} + \theta_A a_{it}$$

then there is an alternative representation

$$\widetilde{v}_{it} - \beta_L l_{it} = \delta k_{it} + \theta_K(\phi_{i,t-1} - \delta k_{i,t-1}) + \xi_{it} + \varepsilon_{it}$$

with $\delta = \beta_K - [(\theta_K - \rho)/\theta_A]$, with the same error term, which is *statistically indistinguishable* from the OP second-stage specification

$$\widetilde{v}_{it} - \beta_L l_{it} = \beta_K k_{it} + \rho(\phi_{i,t-1} - \beta_K k_{i,t-1}) + \xi_{it} + \varepsilon_{it}$$

The variance of $(\xi_{it} + \varepsilon_{it})$ is minimized *both* at the true parameter values (β_K, ρ) *and* at the alternative values (δ, θ_K), and the output elasticity β_K is not identified from the OP second-stage specification

Under the OP assumptions, and if we have a linear AR(1) process for a_{it}, the decision rule for $k_{i,t+1}$ *is* an exact log-linear function of (k_{it}, a_{it}) if

- there are no adjustment costs for capital, or if
- capital is subject to strictly convex adjustment costs which are symmetric (for increases and decreases)

Identification of β_K from the OP second-stage specification then requires that capital is subject to strictly convex adjustment costs which are *asymmetric* (i.e. decreasing the capital stock by $x\%$ is more costly than increasing the capital stock by $x\%$, or vice versa)

Summary

For OP Stage 1 to estimate β_L consistently, we require that l_{it} is a perfectly flexible input, with serially uncorrelated variation across firms in the real wage; we also require that k_{it} is chosen in period $t-1$, with no variation across firms in the price of capital goods, or in other components of the user cost of capital, and either strictly convex or no adjustment costs for capital; and we require a quantity measure of the labour input

For OP Stage 2 to estimate β_K consistently, we further require strictly convex and asymmetric adjustment costs for capital, at least when a_{it} follows a linear AR(1) process

NB. Inference about β_K is *not* standard, since we use a 'generated regressor' $\widehat{\phi}_{i,t-1}$ in OP stage 2

Valid inference can be obtained either analytically (OP 1996) or using a bootstrap procedure (LP 2003)

The validity of the proxy $h_t(k_{i,t+1}, k_{it})$ could be *tested* by adding additional $k_{i,t-1}$ or $l_{i,t-1}$ terms to the Stage 1 specification

$$\widetilde{v}_{it} = \beta_L l_{it} + \phi_t(k_{i,t+1}, k_{it}) + \varepsilon_{it}$$

These lagged inputs should be uninformative if $h_t(k_{i,t+1}, k_{it})$ fully controls for a_{it}, but not if (for example) we have serially correlated variation across firms in either the real wage or the user cost of capital

The Wooldridge (2009) Formulation

Wooldridge (2009) re-formulates the OP procedure as a standard GMM estimator for the system of 2 equations

$$\widetilde{v}_{it} = \beta_K k_{it} + \beta_L l_{it} + h_t(k_{i,t+1}, k_{it}) + \varepsilon_{it}$$

$$\widetilde{v}_{it} = \beta_K k_{it} + \beta_L l_{it} + g(h_{t-1}(k_{it}, k_{i,t-1})) + \xi_{it} + \varepsilon_{it}$$

For the first equation, we have moment conditions

$$E(\varepsilon_{it}|l_{it}, l_{i,t-1}, ..., k_{i,t+1}, k_{it}, k_{i,t-1}, ...) = 0$$

For the second equation, we have moment conditions

$$E(\xi_{it} + \varepsilon_{it}|l_{i,t-1}, l_{i,t-2}, ..., k_{it}, k_{i,t-1}, ...) = 0$$

For example, suppose we specify

$$g(a_{i,t-1}) = \rho a_{i,t-1}$$

for the linear AR(1) case, and

$$h_t(k_{i,t+1}, k_{it}) = \lambda_1 k_{i,t+1} + \lambda_2 k_{it} + \lambda_3 k_{i,t+1}^2 + \lambda_4 k_{it}^2 + \lambda_5 k_{i,t+1} k_{it}$$

as a second-order polynomial

Then we have

$$\widetilde{v}_{it} = \beta_K k_{it} + \beta_L l_{it} + \lambda_1 k_{i,t+1} + \lambda_2 k_{it} + \lambda_3 k_{i,t+1}^2 + \lambda_4 k_{it}^2 + \lambda_5 k_{i,t+1} k_{it} + \varepsilon_{it}$$

$$\widetilde{v}_{it} = \beta_K k_{it} + \beta_L l_{it} + \rho(\lambda_1 k_{it} + \lambda_2 k_{i,t-1} + \lambda_3 k_{it}^2 + \lambda_4 k_{i,t-1}^2 + \lambda_5 k_{it} k_{i,t-1}) + e_{it}$$

where $e_{it} = \xi_{it} + \varepsilon_{it}$

$$\widetilde{v}_{it} = \beta_K k_{it} + \beta_L l_{it} + \lambda_1 k_{i,t+1} + \lambda_2 k_{it} + \lambda_3 k_{i,t+1}^2 + \lambda_4 k_{it}^2 + \lambda_5 k_{i,t+1} k_{it} + \varepsilon_{it}$$

$$\widetilde{v}_{it} = \beta_K k_{it} + \beta_L l_{it} + \rho(\lambda_1 k_{it} + \lambda_2 k_{i,t-1} + \lambda_3 k_{it}^2 + \lambda_4 k_{i,t-1}^2 + \lambda_5 k_{it} k_{i,t-1}) + e_{it}$$

Let $\theta = (\beta_K, \beta_L, \rho, \lambda_1, \lambda_2, \lambda_3, \lambda_4, \lambda_5)$, $u_{it}(\theta) = (\varepsilon_{it}(\theta), e_{it}(\theta))'$ and (for example)

$$Z_{it} = \begin{pmatrix} l_{it} & k_{i,t+1} & k_{it} & k_{i,t+1}^2 & k_{it}^2 & k_{i,t+1}k_{it} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & l_{i,t-1} & k_{it} & k_{i,t-1} & k_{it}^2 & k_{i,t-1}^2 & k_{it}k_{i,t-1} \end{pmatrix}$$

Then we have moment conditions $E[Z_{it}' u_{it}(\theta)] = 0$, and we can estimate θ using a non-linear GMM estimator, for which inference is standard

The Wooldridge (2009) formulation simplifies inference, but requires the same (highly restrictive) assumptions as the original OP estimator for the output elasticities (β_L, β_K) to be estimated consistently

Proxy Methods 2 - Levinsohn-Petrin (2003)

Levinsohn and Petrin (2003) noted that there is compelling evidence to suggest that capital is subject to non-convex forms of adjustment costs - for example, in disaggregated data on investment, we see a non-negligible fraction of (smaller) firms or plants reporting zero annual investment

They considered estimation of a Cobb Douglas production function for (the log of observed) gross output

$$\widetilde{q}_{it} = \beta_K k_{it} + \beta_L l_{it} + \beta_M m_{it} + a_{it} + \varepsilon_{it}$$

where a_{it} is (the log of unobserved) TFP and ε_{it} is an additive measurement error component of $\widetilde{q}_{it}$, which is assumed to be uncorrelated with (k_{is}, l_{is}, m_{is}) for all s, t

$$\widetilde{q}_{it} = \beta_K k_{it} + \beta_L l_{it} + \beta_M m_{it} + a_{it} + \varepsilon_{it}$$

LP discuss estimation of $(\beta_K, \beta_L, \beta_M)$ for the case in which we have quantity data on gross output, capital, labour and intermediate inputs, which we follow here

LP treat both l_{it} and m_{it} as perfectly flexible inputs

Like OP, LP propose to estimate β_L from a first-stage specification that uses a proxy for unobserved a_{it}

They depart from OP by considering intermediate inputs (m_{it}), rather than investment or $k_{i,t+1}$, as the proxy variable

LP's key assumptions are that we can write $m_{it} = m_t(k_{it}, a_{it})$ as a function of the state variables k_{it} and a_{it} *only*, and that $m_t(k_{it}, a_{it})$ is *strictly monotonic* in a_{it}

Inverting $m_{it} = m_t(k_{it}, a_{it})$ to obtain $a_{it} = h_t(k_{it}, m_{it})$, their first-stage specification is then

$$\begin{aligned}
\widetilde{q}_{it} &= \beta_K k_{it} + \beta_L l_{it} + \beta_M m_{it} + a_{it} + \varepsilon_{it} \\
&= \beta_K k_{it} + \beta_L l_{it} + \beta_M m_{it} + h_t(k_{it}, m_{it}) + \varepsilon_{it} \\
&= \beta_L l_{it} + \phi_t(k_{it}, m_{it}) + \varepsilon_{it}
\end{aligned}$$

where $\phi_t(k_{it}, m_{it}) = \beta_K k_{it} + \beta_M m_{it} + h_t(k_{it}, m_{it})$

This looks attractive, but there is a fundamental problem

The assumption that $m_{it} = m_t(k_{it}, a_{it})$ requires that there is no variation across firms in the real price of intermediate inputs *and* no variation across firms in the real wage

But if there is no variation across firms in the real wage, we have no variation in l_{it} given k_{it}, m_{it} and a_{it}, which is needed to identify β_L from this first-stage specification

Note that $m_t(k_{it}, a_{it})$ is *not* the optimality condition for intermediate inputs that we derived previously, which related m_{it} to (k_{it}, l_{it}, a_{it})

However, since LP assume that both intermediate inputs and labour are perfectly flexible inputs, we have 2 such conditions

$$
\begin{aligned}
m_{it} = & \left(\frac{1}{1-\beta_M}\right) \ln \beta_M + \left(\frac{\beta_K}{1-\beta_M}\right) k_{it} + \left(\frac{\beta_L}{1-\beta_M}\right) l_{it} \\
& - \left(\frac{1}{1-\beta_M}\right) (p_{it}^M - p_{it}) + \left(\frac{1}{1-\beta_M}\right) a_{it}
\end{aligned}
$$

and

$$
\begin{aligned}
l_{it} = & \left(\frac{1}{1-\beta_L}\right) \ln \beta_L + \left(\frac{\beta_K}{1-\beta_L}\right) k_{it} + \left(\frac{\beta_M}{1-\beta_L}\right) m_{it} \\
& - \left(\frac{1}{1-\beta_L}\right) (w_{it} - p_{it}) + \left(\frac{1}{1-\beta_L}\right) a_{it}
\end{aligned}
$$

We can combine these conditions to derive

$$
\begin{aligned}
m_{it} = & \left(\frac{(1-\beta_L)\ln\beta_M + \beta_L \ln\beta_L}{1-\beta_M-\beta_L}\right) + \left(\frac{1}{1-\beta_M-\beta_L}\right) a_{it} \\
& + \left(\frac{\left(\frac{\beta_K}{1-\beta_K}\right)(1-\beta_M)(1-\beta_L) + \beta_L\beta_K}{1-\beta_M-\beta_L}\right) k_{it} \\
& - \left(\frac{1-\beta_L}{1-\beta_M-\beta_L}\right)(p_{it}^M - p_{it}) - \left(\frac{\beta_L}{1-\beta_M-\beta_L}\right)(w_{it} - p_{it})
\end{aligned}
$$

With no variation in $(p_{it}^M - p_{it})$ and $(w_{it} - p_{it})$, this has the form $m_{it} = m(k_{it}, a_{it})$, which is a special case of the LP assumption $m_{it} = m_t(k_{it}, a_{it})$

But with no variation in $(w_{it} - p_{it})$, l_{it} is linearly dependent on (k_{it}, m_{it}, a_{it}), and we do not identify β_L from the LP first-stage specification

Nor is this resolved here by considering common variation over time in $(w_{it} - p_{it})$ (and possibly also in $(p_{it}^M - p_{it})$)

This gives us $m_{it} = m_t(k_{it}, a_{it})$, or more explicitly $m_{it} = m(k_{it}, a_{it}, d_t)$, where d_t is a time-specific intercept reflecting the common variation over time in $(w_{it} - p_{it})$

Inverting $m_{it} = m(k_{it}, a_{it}, d_t)$ to obtain $a_{it} = h(k_{it}, m_{it}, d_t)$ then requires time-specific intercepts to be included in the LP first-stage specification, and the variation in l_{it} given (k_{it}, m_{it}, a_{it}) which results from common variation over time in $(w_{it} - p_{it})$ is then not sufficient to identify β_L from the LP first-stage specification

This is shown more generally in Ackerberg, Caves and Frazer (2015), and the implication is that the Levinsohn-Petrin estimator *should not be used*

It should also be clear that β_M is not identified from the LP second-stage specification, which is a variation of the OP second-stage specification, in which the lagged input $m_{i,t-1}$ is used as an instrument for m_{it}, which is correlated with the tfp innovation ξ_{it}

We require no variation across firms in $(p_{it}^M - p_{it})$ in order for the intermediate inputs proxy to be valid

But we would need persistent variation across firms in $(p_{it}^M - p_{it})$ in order for $m_{i,t-1}$ to be an informative instrument for m_{it}

Dynamic Panel Data Methods - Blundell-Bond (2000)

Blundell and Bond (2000) also rely on specifying the dynamic process for the evolution of (the log of) unobserved TFP

The basic idea is to transform the production function so that only the innovation to (the log of) TFP in period t appears in the error term of the transformed specification, and then use current or lagged values of the observed inputs which are uncorrelated with this innovation as instruments for the transformed inputs which appear in this transformed production function

For (low order) linear dynamic processes for tfp = ln(TFP), no proxy for unobserved tfp is required

To illustrate this approach with the same timing assumptions as those used in OP (1996), consider estimation of a Cobb Douglas production function for (the log of observed) value added

$$\widetilde{v}_{it} = \beta_K k_{it} + \beta_L l_{it} + a_{it} + \varepsilon_{it}$$

where a_{it} is (the log of unobserved) TFP and ε_{it} is an additive measurement error component of $\widetilde{v}_{it}$, which is assumed to be uncorrelated with (k_{is}, l_{is}) for all s, t

BB discuss estimation of (β_K, β_L) for the case in which we have quantity data on value added, capital and labour, which we initially follow here

[In their empirical application, they use deflated sales revenue (! more on this later) as the measure of output, a capital stock measure constructed from deflated investment expenditure, and a headcount (number of employees) measure of labour]

We follow OP (1996) in assuming that k_{it} is chosen in period $t-1$, after observing $a_{i,t-1}$ and period $t-1$ prices, but before observing a_{it} and period t prices, so that k_{it} is uncorrelated with the innovation to a_{it} in period t

We follow OP (1996) in assuming that l_{it} is a perfectly flexible input, chosen in period t after observing a_{it} and period t prices, so that l_{it} is correlated with a_{it}, but $l_{i,t-1}$ is uncorrelated with the innovation to a_{it} in period t

[BB (2000) treat both k_{it} and l_{it} as being chosen in period t, possibly subject to adjustment costs, and we will consider this case subsequently]

The key assumption in BB (2000) is that a_{it} is generated by a low order linear ARMA process

To illustrate, suppose that a_{it} follows an AR(1) process

$$a_{it} = \rho a_{i,t-1} + \xi_{it}$$

where ξ_{it} is the innovation in period t, with $E(\xi_{it}|a_{i,t-1}) = 0$

From the production function, we also have

$$a_{it} = \widetilde{v}_{it} - \beta_K k_{it} - \beta_L l_{it} - \varepsilon_{it}$$

$$a_{it} = \rho a_{i,t-1} + \xi_{it}$$

$$a_{it} = \widetilde{v}_{it} - \beta_K k_{it} - \beta_L l_{it} - \varepsilon_{it}$$

Combining these equations, we have

$$(\widetilde{v}_{it} - \beta_K k_{it} - \beta_L l_{it} - \varepsilon_{it}) = \rho(\widetilde{v}_{i,t-1} - \beta_K k_{i,t-1} - \beta_L l_{i,t-1} - \varepsilon_{i,t-1}) + \xi_{it}$$

Or

$$(\widetilde{v}_{it} - \beta_K k_{it} - \beta_L l_{it}) = \rho(\widetilde{v}_{i,t-1} - \beta_K k_{i,t-1} - \beta_L l_{i,t-1}) + \xi_{it} + \varepsilon_{it} - \rho\varepsilon_{i,t-1}$$

Or

$$(\widetilde{v}_{it} - \rho\widetilde{v}_{i,t-1}) = \beta_K(k_{it} - \rho k_{i,t-1}) + \beta_L(l_{it} - \rho l_{i,t-1}) + \xi_{it} + \varepsilon_{it} - \rho\varepsilon_{i,t-1}$$

$$(\widetilde{v}_{it} - \rho\widetilde{v}_{i,t-1}) = \beta_K(k_{it} - \rho k_{i,t-1}) + \beta_L(l_{it} - \rho l_{i,t-1}) + \xi_{it} + \varepsilon_{it} - \rho\varepsilon_{i,t-1}$$

We have a quasi-differenced or dynamic representation of the production function, in which only the tfp innovation (ξ_{it}) appears in the error term, together with the measurement errors in the log of value added in the two periods (ε_{it} and $\varepsilon_{i,t-1}$)

The observed inputs $k_{it}, k_{i,t-1}, ..., l_{i,t-1} l_{i,t-2}, ...$ are uncorrelated with the error term ($\xi_{it} + \varepsilon_{it} - \rho\varepsilon_{i,t-1}$) under the stated assumptions

The observed outputs $\widetilde{v}_{i,t-2}, \widetilde{v}_{i,t-3}, ...$ are also uncorrelated with ($\xi_{it} + \varepsilon_{it} - \rho\varepsilon_{i,t-1}$) if the measurement error ε_{it} is serially uncorrelated

$$(\widetilde{v}_{it} - \rho\widetilde{v}_{i,t-1}) = \beta_K(k_{it} - \rho k_{i,t-1}) + \beta_L(l_{it} - \rho l_{i,t-1}) + \xi_{it} + \varepsilon_{it} - \rho\varepsilon_{i,t-1}$$

Letting $\theta = (\beta_K, \beta_L, \rho)$, $u_{it} = \xi_{it} + \varepsilon_{it} - \rho\varepsilon_{i,t-1}$, and (for example) $Z_{it} = (k_{it}, k_{i,t-1}, l_{i,t-1}l_{i,t-2}, \widetilde{v}_{i,t-2})$, we have moment conditions $E[Z'_{it}u_{it}(\theta)] = 0$, and we can estimate θ using a non-linear GMM estimator, for which inference is standard

These instruments are valid under the stated assumptions, but we also need to consider whether $\theta = (\beta_K, \beta_L, \rho)$ is identified using this approach, i.e. whether these instruments are sufficiently informative

Identification - Flexible Inputs

From our earlier analysis of the optimality condition for perfectly flexible inputs, we know that persistent variation across firms in the (log of the) real wage will be required for lagged inputs $l_{i,t-1}, l_{i,t-2}, ...$ to be informative instruments for $(l_{it} - \rho l_{i,t-1})$ in the quasi-differenced production function

More precisely, here we have

$$l_{it} = \left(\frac{1}{1-\beta_L}\right) \ln \beta_L + \left(\frac{\beta_K}{1-\beta_L}\right) k_{it} - \left(\frac{1}{1-\beta_L}\right) (w_{it} - p_{it}^V) + \left(\frac{1}{1-\beta_L}\right) a_{it}$$

And hence (quasi-differencing)

$$
\begin{aligned}
(l_{it} - \rho l_{i,t-1}) \;=\; & \left(\frac{1-\rho}{1-\beta_L}\right) \ln \beta_L + \left(\frac{\beta_K}{1-\beta_L}\right) (k_{it} - \rho k_{i,t-1}) \\
& - \left(\frac{1}{1-\beta_L}\right) [(w_{it} - p_{it}^V) - \rho(w_{i,t-1} - p_{i,t-1}^V)] + \left(\frac{1}{1-\beta_L}\right) \xi_{it}
\end{aligned}
$$

Given $(k_{it} - \rho k_{i,t-1})$, any informative instrument for $(l_{it} - \rho l_{i,t-1})$ which is uncorrelated with ξ_{it} must be correlated with $[(w_{it} - p_{it}^V) - \rho(w_{i,t-1} - p_{i,t-1}^V)]$

The lagged quasi-difference $(l_{i,t-1} - \rho l_{i,t-2})$ satisfies this requirement only if the (log of the) real wage $(w_{it} - p_{it}^V)$ varies across firms and is serially correlated

Further, recall our result for the cost share of a perfectly flexible input with a Cobb Douglas production function and price-taking firms

Here this becomes

$$\frac{W_{it}L_{it}}{P_{it}^{V}V_{it}} = \beta_L$$

where the left-hand side is the share of labour costs in (correctly measured) value added

Taking logs, we can rearrange this expression to give

$$v_{it} = -\ln \beta_L + l_{it} + (w_{it} - p_{it}^{V})$$

Or with measurement error in the log of value added

$$\widetilde{v}_{it} = -\ln \beta_L + l_{it} + (w_{it} - p_{it}^{V}) + \varepsilon_{it}$$

Suppose that the log of the real wage also follows an AR(1) process

$$(w_{it} - p_{it}^V) = \phi(w_{i,t-1} - p_{i,t-1}^V) + \zeta_{it}$$

Then we have the relation

$$(\widetilde{v}_{it} - \phi\widetilde{v}_{i,t-1}) = -(1-\phi)\ln\beta_L + (l_{it} - \phi l_{i,t-1}) + \zeta_{it} + \varepsilon_{it} - \phi\varepsilon_{i,t-1}$$

Or

$$\begin{aligned}(\widetilde{v}_{it} - \phi\widetilde{v}_{i,t-1}) \;=\; & -(1-\phi)\ln\beta_L + \tau_K(k_{it} - \phi k_{i,t-1}) \\ & +\tau_L(l_{it} - \phi l_{i,t-1}) + \zeta_{it} + \varepsilon_{it} - \phi\varepsilon_{i,t-1}\end{aligned}$$

in which $\tau_K = 0$ and $\tau_L = 1$

Now compare

$$(\widetilde{v}_{it} - \rho\widetilde{v}_{i,t-1}) = \beta_K(k_{it} - \rho k_{i,t-1}) + \beta_L(l_{it} - \rho l_{i,t-1}) + \xi_{it} + \varepsilon_{it} - \rho\varepsilon_{i,t-1}$$

and

$$\begin{aligned}(\widetilde{v}_{it} - \phi\widetilde{v}_{i,t-1}) \;=\; & -(1-\phi)\ln\beta_L + \tau_K(k_{it} - \phi k_{i,t-1}) \\ & +\tau_L(l_{it} - \phi l_{i,t-1}) + \zeta_{it} + \varepsilon_{it} - \phi\varepsilon_{i,t-1}\end{aligned}$$

If the (log of the) real wage does indeed follow an AR(1) process, the same instruments $Z_{it} = (k_{it}, k_{i,t-1}, ..., l_{i,t-1}, l_{i,t-2}, ..., \widetilde{v}_{i,t-2}, \widetilde{v}_{i,t-2}, ...)$ that are orthogonal to the error term $\xi_{it} + \varepsilon_{it} - \rho\varepsilon_{i,t-1}$ in the quasi-differenced production function are also orthogonal to the error term $\zeta_{it} + \varepsilon_{it} - \phi\varepsilon_{i,t-1}$ in the transformed labour demand relation

Given that, these moment conditions do not identify the parameters of interest (β_K, β_L, ρ) in the quasi-differenced production function separately from the parameters (τ_K, τ_L, ϕ) of the transformed labour demand relation

To ensure identification of the production function parameters, we *require* the process for the log of the real wage to have dynamics that are *strictly richer* than the assumed process for the log of TFP

Technically, we require the dynamic process for the log of TFP to be *nested* within the dynamic process for the log of the real wage

- Bond and Söderbom (in preparation)

Maintaining the assumption that we have an AR(1) process for a_{it}, this requires the process for $w_{it} - p_{it}^V$ to have strictly richer dynamics

- for example, AR(2) or ARMA(1,1)

For example, if $w_{it} - p_{it}^V$ follows an AR(2) process, so that ζ_{it} is AR(1), then $l_{i,t-1}$ is not orthogonal to $\zeta_{it} + \varepsilon_{it} - \phi\varepsilon_{i,t-1}$, and hence not a valid instrument in the transformed labour demand relation

Thus if labour is a perfectly flexible input, identification of (β_K, β_L, ρ) from the quasi-differenced production function specification requires that we have persistent variation across producers in the log of the real wage, with strictly richer dynamics than the assumed process for the log of TFP

Identification - Predetermined Inputs

Recall that β_K is not identified from OP Stage 2 in the special case where the decision rule for investment in period t

$$k_{i,t+1} = i_t(k_{it}, a_{it})$$

is an exact log-linear function

$$k_{i,t+1} = \theta_K k_{it} + \theta_A a_{it}$$

In this case, we also have an alternative quasi-differenced relation

$$(\widetilde{v}_{it} - \theta_K \widetilde{v}_{i,t-1}) = \delta(k_{it} - \theta_K k_{i,t-1}) + \beta_L(l_{it} - \theta_K l_{i,t-1}) + \xi_{it} + \varepsilon_{it} - \theta_K \varepsilon_{i,t-1}$$

again with $\delta = \beta_K - [(\theta_K - \rho)/\theta_A]$

$$(\widetilde{v}_{it} - \theta_K \widetilde{v}_{i,t-1}) = \delta(k_{it} - \theta_K k_{i,t-1}) + \beta_L(l_{it} - \theta_K l_{i,t-1}) + \xi_{it} + \varepsilon_{it} - \theta_K \varepsilon_{i,t-1}$$

Here again it is easy to see that the same set of instruments

$$Z_{it} = (k_{it}, k_{i,t-1}, ..., l_{i,t-1}, l_{i,t-2}, ..., \widetilde{v}_{i,t-2}, \widetilde{v}_{i,t-3}, ...)$$

that are orthogonal to the error term $\xi_{it} + \varepsilon_{it} - \rho\varepsilon_{i,t-1}$ in the quasi-differenced production function are also orthogonal to the error term $\xi_{it} + \varepsilon_{it} - \theta_K \varepsilon_{i,t-1}$ in this alternative relation

So that these moment conditions do not identify the parameters of interest (β_K, β_L, ρ) in the quasi-differenced production function separately from the parameters $(\delta, \beta_L, \theta_K)$ of this alternative relation

To ensure identification of the output elasticity with respect to the predetermined input (β_K), we thus *require* that the decision rule for investment is *not* an exact log-linear function of (k_{it}, a_{it}) only

Persistent variation across firms in the log of the real wage is *sufficient* to ensure this, since with *persistent* variation, the optimal investment decision in period t depends on the real wage $(w_{it} - p_{it}^V)$, in addition to k_{it} and a_{it} (with or without any adjustment costs for capital)

Non-convex adjustment costs for capital, such as (partial) irreversibility, would also ensure that the decision rule for investment is not an exact log-linear function of (k_{it}, a_{it}); as would asymmetric strictly convex adjustment costs, or variation across firms in the user cost of capital

Generalizations

Timing assumptions are less important in the context of the BB approach

For example, we could assume that the capital stock used to produce output in period t (k_{it}) is chosen in period t, after the firm has observed a_{it} and period t prices

This implies that k_{it} is correlated with the tfp innovation ξ_{it}, and hence not a valid instrument in the quasi-differenced production function

But $k_{i,t-1}, k_{i,t-2}$ and longer lags are still available as instruments for the quasi-difference $(k_{it} - \rho k_{i,t-1})$ in this case

Similarly, we could assume that labour is subject to adjustment costs (e.g. hiring and firing costs), and hence not a perfectly flexible input

Lagged inputs $l_{i,t-1}, l_{i,t-2}$ and longer lags remain available as instruments for $(l_{it} - \rho l_{i,t-1})$ in this case

Identification requirements with these alternative timing assumptions have not been studied as carefully as for the OP (1996) timing assumptions (to my knowledge), but it seems a reasonable conjecture that persistent variation across firms in the log of the real wage and/or suitable (possibly asymmetric) forms of adjustment costs would also allow the production function parameters (β_K, β_L, ρ) to be identified in these cases

We can also relax the rather arbitrary assumption that only value added is measured with error, and additionally allow for measurement error in one or both of the inputs

Suppose that both k_{it} and l_{it} are subject to additive measurement errors, so that the production function relating observed inputs to observed output is now

$$\widetilde{v}_{it} = \beta_K \widetilde{k}_{it} + \beta_L \widetilde{l}_{it} + a_{it} + \varepsilon^*_{it}$$

where the measurement error ε^*_{it} now combines additive measurement error components of observed $\widetilde{k}_{it}$, $\widetilde{l}_{it}$ and $\widetilde{v}_{it}$

The quasi-differenced production function is now

$$(\widetilde{v}_{it} - \rho\widetilde{v}_{i,t-1}) = \beta_K(\widetilde{k}_{it} - \rho\widetilde{k}_{i,t-1}) + \beta_L(\widetilde{l}_{it} - \rho\widetilde{l}_{i,t-1}) + \xi_{it} + \varepsilon^*_{it} - \rho\varepsilon^*_{i,t-1}$$

Provided these measurement errors are serially uncorrelated and uncorrelated with each other, the instruments

$$Z_{it} = (\widetilde{k}_{i,t-2}, \widetilde{k}_{i,t-3}, ..., \widetilde{l}_{i,t-2}, \widetilde{l}_{i,t-3}, ..., \widetilde{v}_{i,t-2}, \widetilde{v}_{i,t-3}, ...)$$

remain orthogonal to the error term $\xi_{it} + \varepsilon^*_{it} - \rho\varepsilon^*_{i,t-1}$ here

Finally this approach can accommodate unobserved firm-specific 'fixed effects', or a time-invariant, firm-specific component of unobserved tfp

Suppose that the process for a_{it} has the (non-Markov) form

$$a_{it} = a_i + \omega_{it}$$

with

$$\omega_{it} = \rho\omega_{i,t-1} + \xi_{it}$$

The production function is then

$$\widetilde{v}_{it} = \beta_K k_{it} + \beta_L l_{it} + a_i + \omega_{it} + \varepsilon_{it}$$

And the quasi-differenced production function is

$$(\widetilde{v}_{it} - \rho\widetilde{v}_{i,t-1}) = \beta_K(k_{it} - \rho k_{i,t-1}) + \beta_L(l_{it} - \rho l_{i,t-1})$$
$$+ (1-\rho)a_i + \xi_{it} + \varepsilon_{it} - \rho\varepsilon_{i,t-1}$$

Observed inputs and output are correlated with a_i and hence the (lagged) levels of observed inputs and output are not valid instruments in this case

But the time-invariant $(1-\rho)a_i$ component of the error term can be eliminated by first-differencing, and (suitably lagged) levels of observed inputs and output remain orthogonal to the error term $\Delta\xi_{it} + \Delta\varepsilon_{it} - \rho\Delta\varepsilon_{i,t-1}$ after taking first-differences

Moreover, (lagged) first-differences of observed inputs and output may be orthogonal to the error term $(1-\rho)a_i+\xi_{it}+\varepsilon_{it}-\rho\varepsilon_{i,t-1}$ in the quasi-differenced production function, if the observed series $(\widetilde{v}_{it}, k_{it}, l_{it})$ satisfy a stationarity condition

- see BB (2000) for details

Given the high degree of persistence in the observed $(\widetilde{v}_{it}, k_{it}, l_{it})$ series, BB (2000) recommend combining these two sets of moment conditions using a 'system GMM' estimator to allow for firm-specific 'fixed effects' in this context

- see BB (2000) for details

Firm-specific 'fixed effects' also allow for *permanent* measurement error in the observed logs of inputs and output

- e.g. the capital stock for one firm is overstated by 10% in each time period, value-added for another firm is understated by 5% in each time period, ...

The empirical application in BB (2000) assumes that both inputs are chosen in period t, both subject to (unspecified forms of) adjustment costs, allows for firm-specific 'fixed effects', and both serially uncorrelated and permanent measurement error components in the log of value added and in the logs of both inputs

Testing

Since these are standard GMM estimators, with more moment conditions available than the number of parameters to be estimated, the assumptions used to justify the validity of moment conditions for the quasi-differenced specification can be tested using the standard GMM test of overidentifying restrictions, i.e. the Hansen test (or J test)

Particular assumptions required to justify the validity of particular subsets of these moment conditions can also be tested using criterion difference tests, i.e. the Difference Hansen tests (or C tests); or using Hausman test comparisons of parameters estimated with and without the use of these particular moment conditions

The validity of the dynamic specification

$$(\widetilde{v}_{it} - \rho\widetilde{v}_{i,t-1}) = \beta_K(k_{it} - \rho k_{i,t-1}) + \beta_L(l_{it} - \rho l_{i,t-1}) + \xi_{it} + \varepsilon_{it} - \rho\varepsilon_{i,t-1}$$

implied by combining the 'static' Cobb Douglas production function

$$\widetilde{v}_{it} = \beta_K k_{it} + \beta_L l_{it} + a_{it} + \varepsilon_{it}$$

with the assumption of an AR(1) process for a_{it}

$$a_{it} = \rho a_{i,t-1} + \xi_{it}$$

can also be tested

Write the restricted dynamic specification in the form

$$\widetilde{v}_{it} = \rho\widetilde{v}_{i,t-1} + \beta_K k_{it} - \rho\beta_K k_{i,t-1} + \beta_L l_{it} - \rho\beta_L l_{i,t-1} + \xi_{it} + \varepsilon_{it} - \rho\varepsilon_{i,t-1}$$

and compare the unrestricted dynamic specification

$$\widetilde{v}_{it} = \pi_1\widetilde{v}_{i,t-1} + \pi_2 k_{it} + \pi_3 k_{i,t-1} + \pi_4 l_{it} + \pi_5 l_{i,t-1} + \xi_{it} + \varepsilon_{it} - \rho\varepsilon_{i,t-1}$$

If we estimate the unrestricted dynamic model, and our quasi-differenced specification is appropriate, we should not reject the 2 non-linear 'common factor' restrictions $\pi_3 = -\pi_1\pi_2$ and $\pi_5 = -\pi_1\pi_4$

These restrictions can be tested, individually and jointly, using standard Wald tests

Limitations

One limitation of the BB (2000) approach is that it does not extend (straightforwardly) to *non-linear processes* for a_{it}, unless we are willing to assume that both inputs and output are measured without error

To see why, simplify to the case of a single input, denoted by x_{it}, and assume that v_{it} is measured with an additive measurement error (ε_{it})

We have

$$\widetilde{v}_{it} = \beta_X x_{it} + a_{it} + \varepsilon_{it}$$

or

$$a_{it} = \widetilde{v}_{it} - \beta_X x_{it} - \varepsilon_{it}$$

Now suppose that a_{it} is a quadratic function of $a_{i,t-1}$

$$a_{it} = \rho_1 a_{i,t-1} + \rho_2 a_{i,t-1}^2 + \xi_{it}$$

Then we have

$$\begin{aligned}(\widetilde{v}_{it} - \beta_X x_{it} - \varepsilon_{it}) \;=\; & \rho_1(\widetilde{v}_{i,t-1} - \beta_X x_{i,t-1} - \varepsilon_{i,t-1}) \\ & +\rho_2(\widetilde{v}_{i,t-1} - \beta_X x_{i,t-1} - \varepsilon_{i,t-1})^2 + \xi_{it}\end{aligned}$$

With $\rho_2 \neq 0$, this introduces terms of the form $\rho_2 \varepsilon_{i,t-1}^2$, $2\rho_2 \beta_X x_{i,t-1} \varepsilon_{i,t-1}$ and $-2\rho_2 \widetilde{v}_{i,t-1} \varepsilon_{i,t-1}$ into the error term of the transformed production function

More specifically, we have

$$(\widetilde{v}_{it} - \beta_X x_{it}) = \rho_1(\widetilde{v}_{i,t-1} - \beta_X x_{i,t-1}) + \rho_2(\widetilde{v}^2_{i,t-1} + \beta^2_X x^2_{i,t-1} - 2\beta_X \widetilde{v}_{i,t-1} x_{i,t-1})$$

$$+(\xi_{it} + \varepsilon_{it} - \rho_1 \varepsilon_{i,t-1} + \rho_2 \varepsilon^2_{i,t-1} + 2\rho_2 \beta_X x_{i,t-1} \varepsilon_{i,t-1} - 2\rho_2 \widetilde{v}_{i,t-1} \varepsilon_{i,t-1})$$

Finding *any* valid instruments in this case would require *very strong* assumptions on the form of the measurement error (ε_{is}) - for example, homoskedastic, serially independent, and independent of both the observed input (x_{it}) and true value added (v_{it})

- in which case we could basically ignore these additional components of the error term, and use instruments $Z_{it} = (x_{i,t-1}, x_{i,t-2}, ..., \widetilde{v}_{i,t-2}, \widetilde{v}_{i,t-3}, ...)$, as we would if $\rho_2 = 0$ (and x_{it} is correlated with ξ_{it})

NB. Non-linearity in the dynamic relationship between value added and the lagged input(s) $x_{i,t-1}$ could be generated either by a Cobb Douglas production function with a non-linear process for a_{it}, or by a translog production function with a linear process for a_{it}

The BB (2000) approach extends more straightforwardly to the latter case (which also introduces non-linear x_{it}^2 terms into the quasi-differenced specification)

[Although the problem highlighted here also reappears if we find that we require a translog production function with a non-linear process for a_{it}]

Reliance on (persistent) variation across firms in the (log of the) real wage to ensure identification of the production function parameters also requires the use of a *quantity measure* of the labour input (number of employees, or total hours worked)

To see why, assume that all firms use 2 types of workers, skilled and unskilled, possibly in different proportions

Suppose that 1 unskilled worker contributes 1 unit of labour, and 1 skilled worker contributes $1 + \gamma$ (effective) units of labour, for some $\gamma > 0$

A firm that employs L_U unskilled workers and L_S skilled workers thus uses $L_U + L_S(1 + \gamma) \neq L_U + L_S$ (effective) units of labour

Suppose also that the wage for unskilled workers is W_U and the wage for skilled workers is $W_S = W_U(1+\gamma)$, so that the skill premium in the wage reflects the difference in the productivity of the 2 types of workers

The wage bill for a firm which employs L_U unskilled workers and L_S skilled workers is then $W_U L_U + W_S L_S = W_U L_U + W_U(1+\gamma)L_S = W_U[L_U + L_S(1+\gamma)]$

In this case, the wage bill is proportional to the number of (effective) units of labour employed by the firm, and provides us with a quality-adjusted measure of labour input

But this requires that market wages provide the appropriate weights to combine skilled and unskilled workers into a single measure of the (effective) labour input, and that *all firms face the same price* ($W_U = W_S/(1+\gamma)$) for 1 (effective) unit of labour

If we use the wage bill as our quality-adjusted measure of labour input, we cannot then appeal to variation across firms in the (effective) price of labour as the source of identifying variation for estimation of the production function

Of course, it is restrictive to assume that skilled and unskilled workers are perfect substitutes in the production process, apart from this adjustment for the number of effective units of labour that each contributes

But this approach is less restrictive than assuming that all workers are identical, and mirrors what we do when we use expenditure-based measures of capital and intermediate inputs

If we are willing to assume that labour is a perfectly flexible input, we can still use the cost share approach to estimate β_L

- the cost share approach is agnostic about whether or not all firms face the same cost for 1 (effective) unit of labour, so long as the production function is Cobb Douglas and firms are price-takers

Otherwise we require a different source of identifying variation

Identification Without Input Price Variation:

Bond and Söderbom (2005)

We have already seen that the elasticity of output with respect to the predetermined capital input (β_K) can be identified without requiring any variation across firms in the user cost of capital

For example, if we have partial irreversibility so that the decision rule for investment is not an exact log-linear function of (k_{it}, a_{it}), and estimate β_L consistently from the cost share equation, assuming that labour is a perfectly flexible input

We could then estimate β_K consistently from a quasi-differenced version of

$$\widetilde{v}_{it} - \beta_L l_{it} = \beta_K k_{it} + a_{it} + \varepsilon_{it}$$

This is just as well, since in practice we almost always use capital stock measures ($P_{it}^K K_{it}$) which are constructed from investment expenditure ($P_{it}^K I_{it}$) data, and require the price of capital goods (P_{it}^K) to be common across firms to interpret these as if they are quantity measures

Bond and Söderbom (2005) investigate whether we can identify both elasticities (β_K, β_L) from estimation of a quasi-differenced production function when both inputs are subject to adjustment costs, with no variation across firms in either the user cost of capital or in the real wage

The intuition is that while unobserved tfp is Hicks neutral, so that a permanent change in a_{it} has no effect on the capital-labour ratio $(k_{it} - l_{it})$ in the long run, after all input adjustment is completed, this need not be the case in the short run

Differences in adjustment costs will generate different responses of k_{it} and l_{it} to a given change in a_{it} in the short run

For example, with strictly convex adjustment costs, if it is very costly to adjust the capital stock and less costly to adjust employment, then following an increase in a_{it} we would expect to see hiring respond more quickly than investment, generating a reduction in the capital-labour ratio

Bond and Söderbom (2005) investigate whether different adjustment costs alone can generate sufficient variation to identify (β_K, β_L) when all firms face common prices for both inputs

They do so using the same timing assumptions as BB (2000), with k_{it} and l_{it} both chosen in period t, after the firm has observed a_{it}

Even in this case, their simulation results suggest that the answer *may* be yes, although we require forms of adjustment costs which generate asymmetric responses to increases and decreases in a_{it} (to avoid exact log-linearity of the decision rules), and stochastic components of adjustment costs ('adjustment cost shocks') may be needed to obtain useful estimates of the two elasticity parameters in realistic sample sizes

I conjecture that this source of identifying variation would be more useful if we assume instead that k_{it} is chosen in period $t-1$ before the firm has observed a_{it}, so that l_{it} responds to the tfp innovation ξ_{it} in period t, but k_{it} does not, and this innovation affects the evolution of the capital stock only from period $t+1$ onwards

- i.e. using the same timing assumptions as OP (1996), consistent with k_{it} being the measured stock of capital at the beginning of period t

This source of identifying variation could then be crucial in cases where the researcher is not willing to assume that labour is a perfectly flexible input, and uses the wage bill as a quality-adjusted measure of labour input

A *limitation* of this approach is that we require adjustment costs to take the form of *additional financial costs* (e.g. payments to hiring agencies or employment lawyers), not the form of *foregone production* (e.g. some workers who would otherwise have been producing output are diverted to tasks related to hiring or investment)

To allow for adjustment costs in the form of foregone production, we would need to specify the form of the adjustment cost function and estimate this jointly with the production function

Proxy Methods 3: Ackerberg, Caves and Frazer (2015)

The estimation method proposed in ACF (2015) can be viewed either as an extension of the idea in LP (2003), using intermediate inputs (m_{it}) as the proxy for unobserved tfp but avoiding the identification problems with the original LP estimator, or as an extension of the idea in BB (2000), using (lagged) inputs as instruments in a dynamic specification, to handle non-linear processes for tfp

The ACF estimator has two stages, as in OP (1996) and LP (2003), with intermediate inputs used as the proxy for unobserved tfp in the first stage specification, as in LP (2003)

However, unlike in OP (1996) or LP (2003), no parameters of interest are estimated from this first stage specification

The role of the ACF first stage specification is simply to correct the observed log of value added ($\widetilde{v}_{it}$) for an additive measurement error component (ε_{it}), which as we have seen is problematic if we want to use the BB (2000) approach to estimate a specification with a non-linear process for a_{it}

The ACF second stage specification simply applies the BB (2000) approach to estimate the parameters of the production function from the dynamic process specified for a_{it}, using the corrected measure of the log of value added obtained from the first stage

ACF consider estimation of a Cobb Douglas production function for (the log of observed) value added

$$\widetilde{v}_{it} = \beta_K k_{it} + \beta_L l_{it} + a_{it} + \varepsilon_{it}$$

where a_{it} is (the log of unobserved) TFP and ε_{it} is an additive measurement error component of $\widetilde{v}_{it}$, which is assumed to be uncorrelated with (k_{is}, l_{is}) for all s, t

ACF discuss estimation of (β_K, β_L) for the case in which we have quantity data on value added, capital, labour, and intermediate inputs, which we follow here

Their key assumptions are that we can write the level of intermediate inputs $m_{it} = m_t(k_{it}, l_{it}, a_{it})$ as a function of k_{it}, l_{it} and a_{it} *only*, and that $m_t(k_{it}, l_{it}, a_{it})$ is *strictly monotonic* in a_{it}

Note that this is subtly different from LP (2003), who write $m_{it} = m_t(k_{it}, a_{it})$, which requires that l_{it} is a perfectly flexible input

Note also that $m_{it} = m_t(k_{it}, l_{it}, a_{it})$ is *not* the optimality condition for a perfectly flexible input that we derived previously in the context of a Cobb Douglas production function for *gross output* - more on this later

ACF First Stage

Given these assumptions, we can invert $m_{it} = m_t(k_{it}, l_{it}, a_{it})$ to obtain $a_{it} = h_t(k_{it}, l_{it}, m_{it})$ and substitute for a_{it} in the production function

$$\begin{aligned}\widetilde{v}_{it} &= \beta_K k_{it} + \beta_L l_{it} + a_{it} + \varepsilon_{it} \\ &= \beta_K k_{it} + \beta_L l_{it} + h_t(k_{it}, l_{it}, m_{it}) + \varepsilon_{it} \\ &= \phi_t(k_{it}, l_{it}, m_{it}) + \varepsilon_{it}\end{aligned}$$

where $\phi_t(k_{it}, l_{it}, m_{it}) = \beta_K k_{it} + \beta_L l_{it} + h_t(k_{it}, l_{it}, m_{it})$

No parameters of interest are identified from this ACF first stage specification, but since we have assumed that $\widetilde{v}_{it} = v_{it} + \varepsilon_{it}$, we can consider $\widehat{\phi}_{it} = \widehat{\phi}_t(k_{it}, l_{it}, m_{it}) = \widetilde{v}_{it} - \widehat{\varepsilon}_{it}$ to be a consistent estimator of the log of true value added v_{it} under the stated assumptions

ACF Second Stage

Given consistent estimates of $\phi_{it} = \phi_t(k_{it}, l_{it}, m_{it})$ from this first stage specification, the second stage of the ACF procedure uses

$$\begin{aligned} a_{it} &= v_{it} - \beta_K k_{it} - \beta_L l_{it} \\ &= \phi_{it} - \beta_K k_{it} - \beta_L l_{it} \end{aligned}$$

and specifies the dynamic process for the evolution of a_{it}

For example, if we assume a linear AR(1) process $a_{it} = \rho a_{i,t-1} + \xi_{it}$, then we have

$$(\phi_{it} - \beta_K k_{it} - \beta_L l_{it}) = \rho(\phi_{i,t-1} - \beta_K k_{i,t-1} - \beta_L l_{i,t-1}) + \xi_{it}$$

$$(\phi_{it} - \beta_K k_{it} - \beta_L l_{it}) = \rho(\phi_{i,t-1} - \beta_K k_{i,t-1} - \beta_L l_{i,t-1}) + \xi_{it}$$

If we now assume that k_{it} is chosen in period $t-1$, l_{it} is chosen in period t, both inputs possibly subject to adjustment costs, then the set of instruments

$$Z_{it} = (k_{it}, k_{i,t-1}, ..., l_{i,t-1}, l_{i,t-2}, ..., \phi_{i,t-1}, \phi_{i,t-2}, ...)$$

would be orthogonal to the error term ξ_{it} in this dynamic specification

We can estimate the parameters (β_K, β_L, ρ) consistently using these moment conditions, plugging in consistent estimates of ϕ_{it} obtained from the first stage

- provided that (β_K, β_L, ρ) is identified here

That is, provided that we have sufficiently persistent variation across firms in (the log of) the real wage and/or asymmetric adjustment costs, to rule out the case of exact log-linear decision rules

Note that inference is not standard here, since we use a 'generated regressor' $\widehat{\phi}_{it}$ in ACF stage 2

Valid inference can be obtained using a bootstrap procedure (ACF 2015)

For the linear AR(1) tfp process, there is a trade-off between using the ACF estimator based on

$$(\phi_{it} - \beta_K k_{it} - \beta_L l_{it}) = \rho(\phi_{i,t-1} - \beta_K k_{i,t-1} - \beta_L l_{i,t-1}) + \xi_{it}$$

with instruments

$$Z_{it} = (k_{it}, k_{i,t-1}, ..., l_{i,t-1}, l_{i,t-2}, ..., \phi_{i,t-1}, \phi_{i,t-2}, ...)$$

Or using the BB estimator based on

$$(\widetilde{v}_{it} - \beta_K k_{it} - \beta_L l_{it}) = \rho(\widetilde{v}_{i,t-1} - \beta_K k_{i,t-1} - \beta_L l_{i,t-1}) + \xi_{it} + \varepsilon_{it} - \rho\varepsilon_{i,t-1}$$

with instruments (under the same timing assumptions)

$$Z_{it} = (k_{it}, k_{i,t-1}, ..., l_{i,t-1}, l_{i,t-2}, ..., \widetilde{v}_{i,t-2}, \widetilde{v}_{i,t-3}, ...)$$

With measurement error in (the log of) value added, the BB estimator requires that this measurement error is serially uncorrelated, and uses longer lagged values of measured value added in the instrument set; but does not require the validity of any proxy for unobserved tfp, and inference is standard

The main advantage of the ACF estimator is that it extends straightforwardly to non-linear dynamic processes for a_{it}, provided the proxy used for unobserved tfp in the first stage specification is valid

Testing

The validity of the proxy $h_t(k_{it}, l_{it}, m_{it})$ could be tested by adding additional $k_{i,t-1}$, $l_{i,t-1}$ or $m_{i,t-1}$ terms to the first stage specification

$$\widetilde{v}_{it} = \phi_t(k_{it}, l_{it}, m_{it}) + \varepsilon_{it}$$

These lagged inputs should be uninformative if $h_t(k_{it}, l_{it}, m_{it})$ fully controls for a_{it}

Results obtained using the ACF estimator should be viewed with suspicion if one or more of these lagged inputs is found to be significant in this extended first stage specification

Limitations

ACF (2015), p2428, emphasize that the ACF estimator should *only* be used to estimate *value added* production functions

To apply this approach to estimate a gross output production function, we would require $m_{it} = m_t(k_{it}, l_{it}, a_{it})$ to be the optimality condition for a perfectly flexible input, which rules out variation across firms in the real price of intermediate inputs

But then we would have no informative instruments for (quasi-differenced) m_{it} in the second stage specification

The leading case in which the *existence* of a value added production function can be rationalized assumes that the technology for producing gross output is **Leontief** in value added and intermediate inputs

- this is the case discussed in ACF (2015), p2428, and the case considered in the DGP for their Monte Carlo simulations (p2437)

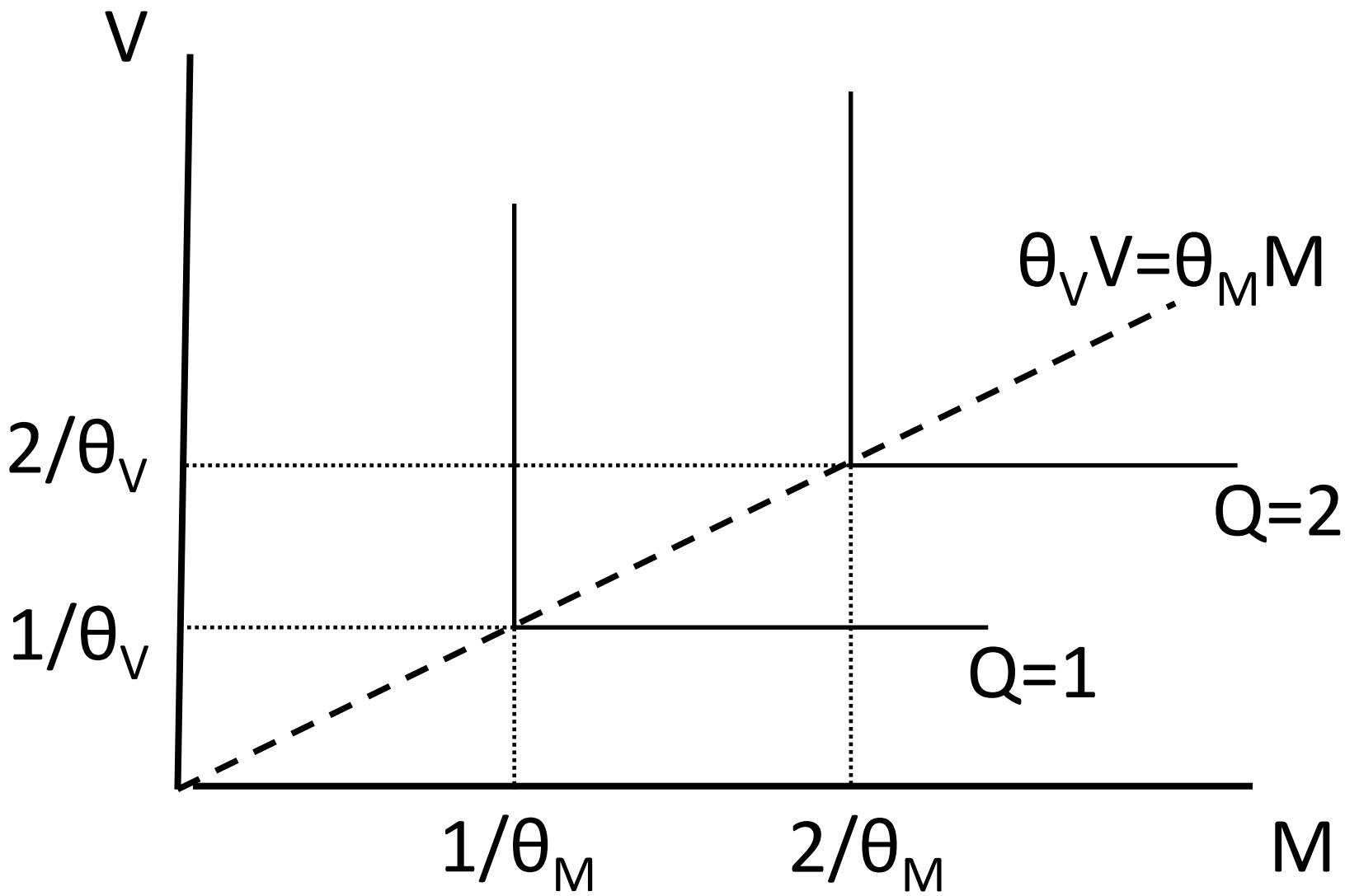
That is, we have

$$Q_{it} = \min(\theta_V V_{it}, \theta_M M_{it})$$

where $V_{it} = v_t(K_{it}, L_{it}, A_{it})$ is value added

Cost minimising behaviour then implies that V_{it} and M_{it} are combined in **fixed proportions** to produce Q_{it}

Gross Output Isoquants



Cost minimisation implies

$$Q_{it} = \theta_V V_{it} = \theta_M M_{it} \quad \text{or} \quad \frac{V_{it}}{M_{it}} = \frac{\theta_M}{\theta_V} = \theta$$

Taking this specification seriously leads to some interesting implications

Taking logs, we have

$$m_{it} = -\ln\theta + v_{it} = -\ln\theta + \ln v_t(K_{it}, L_{it}, A_{it})$$

so the Leontief specification already *implies* the key ACF assumption that

$m_{it} = m_t(k_{it}, l_{it}, a_{it})$

Rearranging, we have

$$v_{it} = \ln \theta + m_{it}$$

With additive measurement error, this becomes

$$\widetilde{v}_{it} = \ln \theta + m_{it} + \varepsilon_{it}$$

Compare the ACF first stage specification

$$\widetilde{v}_{it} = \phi_t(k_{it}, l_{it}, m_{it}) + \varepsilon_{it}$$

The function $\phi_t(k_{it}, l_{it}, m_{it})$ takes a particularly simple form in this case, with an intercept, a coefficient of 1 on m_{it}, and coefficients of zero on *any* other explanatory variables that we include

Results from estimation of the ACF first stage specification can be checked for conformity with these restrictions

If we reject these restrictions, this may indicate that m_{it} is measured with error - in which case measured m_{it} is not a valid proxy for unobserved tfp

Or this may indicate that the technology is not Leontief in value added and intermediate inputs - in which case it is not clear that we should proceed to estimate a value added production function

If we do not reject these restrictions, this also implies that the first stage of the ACF procedure is redundant here

Recall that the purpose of the first stage is to purge the observed log of value added $\widetilde{v}_{it}$ of its measurement error component ε_{it}

But if we have $v_{it} = \ln\theta + m_{it}$, we can measure the true log of value added, up to an additive constant, using the log of intermediate inputs

In this case, we can proceed directly to the second stage of the ACF procedure, replacing

$$a_{it} = v_{it} - \beta_K k_{it} - \beta_L l_{it}$$

by

$$a_{it} = m_{it} + \ln\theta - \beta_K k_{it} - \beta_L l_{it}$$

And using the quasi-differenced specification

$$(m_{it} + \ln\theta - \beta_K k_{it} - \beta_L l_{it}) = \rho(m_{i,t-1} + \ln\theta - \beta_K k_{i,t-1} - \beta_L l_{i,t-1}) + \xi_{it}$$

with instruments

$$Z_{it} = (k_{it}, k_{i,t-1}, ..., l_{i,t-1}, l_{i,t-2}, ..., m_{i,t-1}, m_{i,t-2}, ...)$$

That is, we can use m_{it} directly as the dependent variable in our value added production function, provided that we include a constant

Inference is standard here, as no generated regressors are used in this specification

Since there is no measurement error component in the error term, this specification can also easily be extended to allow for non-linear dynamic processes for a_{it}

These results all extend to the more realistic case in which we use data on the value of gross output (or sales revenue) and expenditure on intermediate inputs, deflated using sector-specific price indices, to construct a measure of value added in constant prices

- provided the prices of gross output, intermediate inputs and value added are common to all firms in the sector, so that we can interpret these deflated revenue and expenditure measures as if they are quantity measures

Aside

For the Leontief case, we also have $Q_{it} = \theta_V V_{it}$ or $q_{it} = \ln \theta_V + v_{it}$

In this case, we can estimate the Cobb Douglas production function for value added using either v_{it} or q_{it} as the dependent variable, provided that we include a constant

- what gives us a value added production function interpretation is the exclusion of intermediate inputs from the explanatory variables considered, not the choice of the dependent variable

Specifications which relate q_{it} to (k_{it}, l_{it}) are very common in empirical work using company accounts data for US firms, which typically report sales revenue, investment expenditure and employment, but report insufficient information to construct a measure of value added

- for example, BB (2000)